

MECH 191 — Metallurgy: Key Terms & Vocabulary Guide

Terms to notate during lecture and research further — organized by unit and week

MECH 191 — Metallurgy

Leeward Community College · Math and Science Division · Mechatronics Program

Compiled from course lecture slides and reference pages

How to use this guide

This list follows the course roadmap — five units, each spanning several weeks. For every week, the terms below are the vocabulary worth writing down during lecture (on your note-taking handout) and looking up further if the in-class explanation leaves you wanting more depth. A short working definition is included for context, but the real value is in researching each term further: find a diagram, a real-world failure example, or the relevant code/standard that governs it.

A few terms repeat across weeks (e.g., grain boundaries, sensitization) — that repetition is intentional in the course design, since Unit 5 explicitly reconnects earlier concepts to failure and corrosion.

A "Units of Measurement & Dimensional Analysis" section comes next — it starts from the seven SI base units, shows how dimensional analysis builds every derived unit (force, stress, energy, power) used in this course from just length, mass, time, and temperature, walks through a few worked examples checking the units a formula produces, and closes with the U.S. customary units (ksi, ft-lbf, °F, mils) still used alongside SI on the shop floor.

A "Quick Math Review" follows that — order of operations, fractions and ratios, percentages, exponents, square roots, scientific notation and log scales, and rearranging an equation to solve for an unknown — each rule illustrated with a worked example pulled directly from a formula elsewhere in this guide (the carbon equivalent formula, Hall-Petch, the Taylor tool life equation, the Brinell hardness formula, and others).

A short "Geometry & Right-Triangle Trigonometry" refresher comes next — angles, circle area and circumference, the Pythagorean theorem, and the sine/cosine/tangent ratios (SOH-CAH-TOA) — grounded in two real course calculations: the 0.707 fillet weld throat factor and the depth/surface-distance trig used to locate a flaw with an angle-beam ultrasonic probe.

A final section, "Engineering Equations & Key Concepts," collects the formulas used across the course, grouped by topic rather than by week since they resurface throughout the semester: stress/strain and true stress-strain relationships, stress concentration and notch effects (K_t , K_f , notch sensitivity), hardness number calculations, fatigue and impact energy, manufacturing equations for cutting mechanics and tool life, forming and sheet-metal relationships (extrusion ratio, forming limit diagrams), weldability indices (CE, Pcm), corrosion (PREN), and NDT physics.

Unit 1 — Materials Fundamentals

Week 1 — Atomic Structure & Bonding

Metallurgy	The science of metals — structure, properties, and behavior.
Alloy	An intentional mixture of two or more elements engineered for desired properties.
Proton / Neutron / Electron	The three subatomic particles; protons define the element, neutrons define isotopes, electrons govern bonding.
Valence electrons	Outermost electrons — the only ones involved in bonding.
Periodic table groups & periods	Columns (groups) share bonding behavior; rows (periods) add electron shells.
Metallic bond	Electrons form a shared 'electron sea' — explains conductivity and ductility.
Ionic bond	Electron transfer between metal and non-metal; high melting point, brittle.
Covalent bond	Shared electrons between non-metals; strong, directional bonds.
Octet rule	Atoms bond to reach 8 stable valence electrons.
Substitutional solid solution	Alloying atoms of similar size replace host atoms in the lattice.
Interstitial solid solution	Small alloying atoms fit into gaps between host atoms (e.g., carbon in iron).
Unit cell	Smallest repeating building block of a crystal structure.
SC / BCC / FCC / HCP	Simple cubic, body-centered cubic, face-centered cubic, hexagonal close-packed crystal structures — know packing % and ductility for each.
Grain / grain boundary	A single crystal domain; the boundary between differently oriented grains.
Polycrystalline	Metal composed of many grains with different orientations.

Week 2 — Mechanical Properties & Behavior

Stress (σ) / Strain (ϵ)	Internal force per unit area; deformation relative to original size.
Tensile / compressive / shear stress	The three basic loading types.
Elastic deformation vs. plastic deformation	Recoverable vs. permanent deformation.
Yield point / yield strength	Stress at onset of permanent plastic deformation.
Ultimate tensile strength (UTS)	Maximum stress on the stress-strain curve, where necking begins.
Strength, Hardness,	The five key mechanical properties technicians must compare.

Ductility, Toughness, Brittleness	
Point defect: vacancy, interstitial, substitutional	Missing atom; extra squeezed-in atom; wrong atom in a lattice site.
Dislocation (edge, screw, mixed)	Line defects that let metal deform plastically; described by the Burgers vector.
Grain boundary (planar defect)	Region where crystal orientation changes; high vs. low misorientation.
Twin boundary (annealing vs. deformation twins)	Mirror-image boundary; low energy, very stable.
Stacking fault	Disruption in the normal atomic stacking sequence; relates to stacking fault energy (SFE).
Voids/porosity, inclusions, delaminations/segregation, cracks	The four volume-defect categories.
Hall-Petch relationship	$\sigma_y = \sigma_0 + k/\sqrt{d}$ — smaller grain size increases yield strength.
Recovery, recrystallization, grain growth	The three stages of annealing after cold work.
Diffusion (substitutional vs. interstitial)	Atomic movement through a solid; rate increases exponentially with temperature.
Work hardening	Strengthening from dislocation accumulation during cold work.
Fatigue / endurance limit	Failure below yield strength from repeated cyclic loading.

Unit 2 — Steel & Alloys

Week 4 — Carbon & Alloy Steels

Ferrite / austenite (BCC/FCC iron)	Low-temp BCC iron phase vs. high-temp FCC iron phase — the basis of steel allotropy.
Carbon equivalent (CE)	$CE = \%C + \%Mn/6 + (\%Cr + \%Mo + \%V)/5 + (\%Ni + \%Cu)/15$ — determines weldability/preheat needs.
Low / medium / high carbon steel	Carbon content ranges and how each affects hardness, ductility, weldability.
AISI/SAE four-digit designation	First two digits = steel family/alloy; last two = carbon content in hundredths of a percent.
Alloying elements: Mn, Cr, Ni, Mo, V, Si	Research what property each element primarily contributes to steel.
Hardenability	How deeply a steel can be hardened through its cross-section (different from

	hardness).
CCT diagram (Continuous Cooling Transformation)	Plots temperature vs. time to show what microstructure actually forms during real cooling.
Ms / Mf (martensite start/finish)	Temperatures at which martensite begins and finishes forming.
Pearlite, Bainite, Martensite	The three microstructures steel can form depending on cooling rate.
Annealing	Heat to austenite, furnace cool — softens and restores ductility.
Normalizing	Heat to austenite, air cool — refines grain size.
Quenching	Rapid cool from austenite — maximizes hardness (forms martensite).
Tempering	Reheating quenched martensite — restores toughness, reduces hardness/brittleness.
Case hardening (carburizing / nitriding)	Hard surface layer over a tough core.
Post-weld heat treatment (PWHT)	Heat treatment after welding to relieve stress and temper the HAZ.

Week 5 — Stainless Steels & Cast Irons

Passive chromium oxide layer (Cr_2O_3)	Self-repairing surface film that makes stainless steel corrosion-resistant; requires $\geq 10.5\%$ Cr.
Austenitic stainless (300 series)	FCC, non-magnetic, not hardenable by heat treatment — e.g. 304, 316.
Ferritic stainless (400 series)	BCC, magnetic, not hardenable — e.g. 430.
Martensitic stainless (400 series)	BCC/BCT, magnetic, hardenable by quench & temper — e.g. 410, 420, 440C.
Duplex stainless	Mixed austenite + ferrite microstructure — higher strength, better SCC resistance.
Sensitization	Chromium carbide precipitation at grain boundaries during welding — depletes nearby Cr, causing intergranular corrosion susceptibility.
L-grade (304L, 316L)	Low-carbon grades used to prevent sensitization in welded fabrication.
Pitting / crevice corrosion / stress corrosion cracking (SCC)	Localized corrosion mechanisms especially relevant to stainless in chloride environments.
Cast iron (definition)	Iron-carbon alloy $> 2.14\%$ carbon, typically 2.5-4%, plus 1-3% silicon.
Gray cast iron	Carbon as graphite flakes — good damping/machinability, low tensile strength.

White cast iron	Carbon as iron carbide/cementite — extremely hard, brittle, wear-resistant.
Ductile (nodular) iron	Mg/Ce addition makes graphite form as spheroids — much tougher than gray iron.
Malleable cast iron	White iron annealed to convert cementite to temper carbon rosettes.
Cementite (Fe₃C)	Iron carbide — extremely hard, brittle phase formed above carbon solubility limit.

Week 6 — Nonferrous Metals & Phase Diagrams

Aluminum alloy series (1xxx–7xxx)	Designation system by principal alloying element; heat-treatable vs. work-hardened series.
T6 temper	Solution treat + age — most common heat-treated condition for 6061/7075 aluminum.
Brass (Cu-Zn) vs. Bronze (Cu-Sn / Cu-Al)	The two major copper alloy families — know composition and typical uses.
Alpha, beta, alpha-beta titanium alloys	HCP, BCC, and mixed titanium microstructures; Ti-6Al-4V is the workhorse alpha-beta alloy.
Iron-carbon phase diagram	Maps stable phases across carbon content and temperature — ferrite, austenite, cementite, pearlite.
Eutectoid point	0.77% C at 727°C — austenite transforms simultaneously to ferrite + cementite.
Hypoeutectoid / eutectoid / hypereutectoid steel	Steel classification by carbon content relative to the eutectoid point (0.77% C).
Galvanic risk / anodic-cathodic relationship (Al vs. steel/Cu)	Aluminum is anodic to steel and copper — avoid direct contact in wet environments.

Unit 3 — Testing, Inspection & Quality

Week 7 — Mechanical Testing Methods

0.2% offset method	Technique for finding yield strength when no clear yield point exists (e.g., aluminum).
Reduction in area / elongation at fracture	Ductility measures from a tensile test.
Rockwell (HRC/HRB), Brinell (HB), Vickers (HV) hardness	Three hardness test methods — different indenters, loads, and best-use cases.
Charpy impact test / Izod	Notched-specimen impact tests measuring absorbed energy (toughness).

test	
Ductile-brittle transition temperature (DBTT)	Temperature below which a material's fracture behavior shifts from tough to brittle.
Upper shelf / lower shelf energy	High-energy ductile fracture region vs. low-energy brittle fracture region on an impact-energy-vs-temperature curve.
S-N curve (Wöhler curve)	Plots stress amplitude vs. cycles to failure — used to find the fatigue limit.
Fatigue limit / fatigue ratio	Stress below which steel can survive indefinite cycles; ratio of fatigue limit to UTS.
Mill Test Report (MTR)	Traceability document linking a heat/lot number to material grade, composition, and mechanical test results.
Heat number / lot number	Unique identifier tying a physical piece of material to its test certificate.

Week 8 — Non-Destructive Testing (NDT) Methods

Visual testing (VT)	The foundational inspection method — naked eye, lighting, magnification, borescopes.
Liquid/dye penetrant testing (PT)	Capillary action draws penetrant into surface-breaking defects; color contrast vs. fluorescent.
Magnetic particle testing (MT)	Detects surface/near-surface defects in ferromagnetic materials via flux leakage; yoke vs. prod methods.
Ultrasonic testing (UT)	Sound waves detect internal defects; A-scan, phased array (PAUT), straight vs. angle beam.
Radiographic testing (RT)	X-ray/gamma imaging of volumetric defects; film vs. digital radiography.
Eddy current testing (ET)	Induces circulating currents to detect surface cracks/corrosion; governed by skin depth.
Couplant	Gel/oil/water medium between UT transducer and part surface (air poorly transmits ultrasound).
Dwell time	How long penetrant is left on the surface before removal in PT.
Two-direction rule (MT)	Inspections must be performed in at least two perpendicular directions.
ASNT SNT-TC-1A Level I/II/III	NDT technician qualification levels — perform, interpret/accept-reject, and write procedures/certify.

Week 9 — Quality Standards & Inspection

ASTM, AWS, ASME, ISO, API	The major standards bodies — know what each governs (materials, welding codes, pressure vessels, international, oil & gas).
Indication → Discontinuity → Defect → Reject	The precise evaluation sequence — memorize the distinct legal meaning of each term.

Porosity / slag inclusion / lack of fusion (LOF) / lack of penetration (LOP)	The primary weld manufacturing defects — know cause and appearance for each.
Undercut / overlap	Weld toe groove vs. rolled-over unfused weld metal — both detectable by VT.
Stress corrosion cracking (SCC) / creep damage	Service-induced defect mechanisms distinct from manufacturing defects.
Non-Conformance Report (NCR)	Formal documentation raised when a component is rejected.
Traceability	Linking an inspection record to the physical component via unique identifiers.

Unit 4 — Manufacturing Processes

Week 10 — Casting Processes & Defect Recognition

Solidification zones (chill, columnar, equiaxed)	The three regions of a solidifying casting, from mold wall inward.
Dendrite / dendrite arm spacing (DAS)	Tree-like solidification structure; finer DAS = better mechanical properties.
Riser (feeder)	Reservoir of liquid metal that compensates for solidification shrinkage.
Sand casting	Pattern, mold, cores, gating system — flexible, low tooling cost, rough finish.
Die casting (hot chamber vs. cold chamber)	High-pressure injection into a reusable die — limited to lower-melting alloys.
Investment (lost-wax) casting	Ceramic shell over a wax pattern — best surface finish and accuracy of any casting process.
Shrinkage porosity / gas porosity	Two major casting defect types — irregular voids vs. smooth spherical voids.
Hot tear / cold shut / misrun	Casting defects from mold restraint, cold metal streams not fusing, and incomplete filling.
Gating system (sprue, runner, gates)	Channels that carry molten metal into the mold cavity.

Week 11 — Metal Forming Processes

Recrystallization temperature (T_{rex})	The boundary (~0.4 T _m) between hot working and cold working.
Hot working vs. cold	No work hardening vs. work hardening — know the trade-offs of each.

working	
Anisotropy	Directional variation in mechanical properties caused by cold-worked crystallographic texture.
Strain aging	Interstitial atoms pinning dislocations after cold working, raising yield strength (Lüders bands).
Rolling (hot/cold, shape rolling)	Reducing thickness between rolls — plate, sheet, structural shapes.
Forging (open die vs. closed/impression die)	Compressive shaping process producing the highest-integrity parts.
Grain flow	Elongated, aligned grains following the part's metal-flow direction in forgings — improves fatigue life.
Flash	Excess metal squeezed out at the die parting line in closed-die forging.
Extrusion (direct vs. indirect)	Forcing metal through a die to produce constant-cross-section profiles.
Drawing (wire, tube, deep drawing)	Pulling metal through a die or punch under tension.
Laps, seams, bursts, pipe defect, chevron/center burst cracking	Characteristic forming defects — know cause, location, and detection method for each.

Week 12 — Machining

Shear plane / primary, secondary, tertiary deformation zones	Where chip formation, tool-chip friction, and flank rubbing occur during cutting.
Continuous vs. discontinuous chip	Chip type depending on material ductility and cutting parameters.
Built-up edge (BUE)	Workpiece material welding to the tool at low cutting speed, causing a rough finish.
Cutting speed (Vc), feed (f), depth of cut (ap)	The three fundamental machining parameters.
Turning, milling, drilling, grinding	The core machining processes — know what surface/feature each produces.
Up milling vs. down (climb) milling	Chip-thickness direction and resulting surface finish/residual stress differences.
Cutting tool materials: HSS, carbide, ceramics, CBN, PCD	Tool material hierarchy by hardness, heat resistance, and application.
Surface roughness: Ra and Rz	Arithmetic mean roughness vs. peak-to-valley height — the two key surface finish parameters.

Grinding burn (re-tempering vs. re-hardening burn)	Thermal damage from excessive grinding heat — detected by Nital etch.
Chatter marks	Vibration-induced wave pattern on a machined surface.
Recast layer (EDM)	Re-solidified surface layer from electrical discharge machining containing tensile residual stress.

Week 13 — Powder Metallurgy

Water atomization vs. gas atomization	Powder production methods producing irregular vs. spherical particle shapes.
Green compact / green strength	The fragile, pressed-but-unsintered powder part before sintering.
Sintering	Densification by solid-state atomic diffusion below the melting point — not melting.
Liquid phase sintering	A lower-melting additive melts and fills pores by capillary action (key to WC-Co tools).
Hot isostatic pressing (HIP)	Simultaneous heat + isostatic gas pressure — eliminates porosity, approaches wrought properties.
Cold isostatic pressing (CIP)	Room-temperature isostatic pressing — eliminates density gradients in complex shapes.
Metal injection molding (MIM)	Fine powder + polymer binder injection molded like plastic, then debound and sintered.
Density gradient / lamination cracks / blistering	The primary PM-specific defects and their causes.
Additive manufacturing (SLM, EBM, DED)	Layer-by-layer metal powder fusion processes — no tooling required.

Unit 5 — Joining, Failure & Corrosion

Week 14 — Welding Processes

SMAW (Shielded Metal Arc Welding / Stick)	Coated consumable electrode; flux forms shielding gas and slag.
GMAW (Gas Metal Arc Welding / MIG)	Continuous bare wire with shielding gas; transfer modes: short circuit, spray, pulsed.
GTAW (Gas Tungsten Arc Welding / TIG)	Non-consumable tungsten electrode; highest quality; AC for aluminum, DCEN for steel.

FCAW (Flux-Cored Arc Welding)	Tubular flux-filled wire; self-shielded (FCAW-S) vs. gas-shielded (FCAW-G).
Heat input formula	$HI = (V \times A \times 60) / (1000 \times \text{travel speed}) \rightarrow \text{kJ/mm}$ — controls cooling rate and grain size.
AWS filler metal classification (e.g., E7018, ER308L, ER4043)	Decode prefix, strength, position, and flux/chemistry codes.
Preheat / interpass temperature	Minimum temperature before/during welding to slow cooling and prevent HAZ cracking.
Carbon equivalent (CE) and preheat thresholds	CE determines required preheat — memorize the CE bands (<0.40, 0.40–0.60, >0.60).
Weld symbol anatomy (reference line, arrow/other side, tail)	How to read the AWS A2.4 weld symbol standard.
Joint types: butt, T-joint, lap, corner, edge	The five basic weld joint configurations.
Groove preparations: square, V, U, bevel	Edge preparation types and when each is used.

Week 15 — Weld Metallurgy & Failure

Heat-affected zone (HAZ)	Base metal heated above ~730°C but not melted — the most failure-prone weld region.
CGHAZ / FGHAZ / ICHAZ / SCHAZ	The four HAZ sub-zones by peak temperature — coarse-grained, fine-grained, intercritical, subcritical.
Acicular ferrite / allotriomorphic ferrite	Weld metal microstructures — the desirable fine interlocking plates vs. the crack-prone grain-boundary phase.
Dilution	The proportion of melted base metal mixed into the deposited weld metal.
Hot cracking (solidification cracking)	Cracking from low-melting films at grain boundaries during solidification; controlled by Mn:S ratio.
Hydrogen-induced cracking (HIC) / cold cracking	Delayed cracking requiring three simultaneous conditions: susceptible microstructure, hydrogen, tensile stress.
Lamellar tearing	Through-thickness base-metal cracking from elongated MnS inclusions under weld shrinkage stress; prevented with Z-quality steel.
Ductile fracture (microvoid coalescence, cup-and-cone)	Fracture with plastic deformation, dimpled surface.
Brittle fracture (cleavage, chevron marks, river	Sudden fracture with no plastic deformation; flat, crystalline surface.

marks)	
Fatigue fracture (beach marks, striations, ratchet marks)	Progressive fracture from cyclic loading; beach marks visible to the eye, striations only under SEM.
Failure analysis methodology	Preserve → document → examine → NDT → mechanical/metallographic test → root cause → report.

Week 16 — Corrosion & Course Wrap-up

Four-element corrosion cell (anode, cathode, electrolyte, metallic path)	All four must be present for electrochemical corrosion — remove one to stop it.
Galvanic series	Ranking of metals by electrochemical potential — determines which metal corrodes when dissimilar metals contact.
Passivation	Formation of a stable, adherent oxide layer that slows corrosion (stainless, aluminum, titanium).
Uniform (general) corrosion	Even, predictable surface loss measured in mm/year or mils per year (MPY).
Galvanic corrosion / area effect	Accelerated corrosion of the less-noble metal when coupled to a more-noble one; area ratio matters.
Pitting corrosion / PREN	Localized passive-film breakdown; Pitting Resistance Equivalent Number quantifies alloy resistance.
Crevice corrosion	Localized attack in stagnant, oxygen-depleted gaps (gaskets, lap joints).
Stress corrosion cracking (SCC)	Tensile stress + specific corrosive environment causes cracking below yield strength.
Cathodic protection (sacrificial anode vs. impressed current)	Makes the structure the cathode so corrosion occurs only at a sacrificial anode.
Hot-dip galvanizing / thermal spray / anodizing	Common protective coating methods — know the mechanism of each.
Corrosion under insulation (CUI)	Hidden corrosion beneath insulation, detected by pulsed eddy current or radiographic profile.

Reference Material — Designation Systems & Field Skills

Steel, Aluminum, Copper & Titanium Designations

AISI/SAE steel series (10xx, 41xx, 43xx, 51xx, 52100, etc.)	Research the alloy family meaning of each major series prefix.
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Aluminum Association temper codes (O, H, T4, T6, T651)	The letter-number suffix after an aluminum alloy number — determines actual strength/hardness.
UNS copper numbering (Cxxxxx) / CDA	Unified Numbering System blocks for coppers, brasses, bronzes, and copper-nickels.
ASTM titanium grades (Grade 2, Grade 5/Ti-6Al-4V, Grade 23/ELI)	Grade number carries no composition meaning by itself — sort into CP/alpha, near-alpha, alpha-beta, beta families.
Extra Low Interstitial (ELI)	Titanium grade variant with tighter oxygen/interstitial control for higher ductility and fracture toughness.

Quality Codes, PPE & Shop Practice

AWS D1.1 (Structural Welding Code — Steel)	Governing code for welded steel structures; know its major clause structure.
ASME BPVC (Boiler and Pressure Vessel Code)	Governs pressure-critical equipment design and fabrication.
Welding Procedure Specification (WPS)	Document specifying process, filler metal, preheat, and parameters for a qualified weld.
Prequalified joint / procedure qualification	Joint details that don't require separate procedure testing under a code.
ANSI Z87.1 / Z87+	Eye protection rating standards for basic vs. high-impact hazards.
Sensitization vs. passivation (shop context)	Reinforce these terms in the context of fabrication practice, not just theory.

Units of Measurement & Dimensional Analysis

The SI Base Units

Base unit (concept)	One of seven independently defined units from which every other physical quantity's unit is built by multiplication and division. None of the seven can be expressed in terms of the others.
Length — meter (m)	The base unit for size, distance, diameter, thickness, and depth. This course mixes it constantly with millimeters (mm) and, in U.S. shop practice, inches (in).
Mass — kilogram (kg)	The base unit for the amount of matter in an object — distinct from weight (a force). Alloy compositions are reported in mass % (wt%), not the mole-based at% used in some materials science contexts.
Time — second (s)	The base unit underlying every rate: cooling rate (°C/s), cutting speed (m/min), frequency (Hz = 1/s), corrosion rate (mm/year).

Temperature — kelvin (K)	The SI base unit; this course reports working temperatures in Celsius (°C) almost exclusively, with Fahrenheit (°F) appearing in AWS D1.1 preheat tables and some U.S. industry references.
Electric current — ampere (A)	The base unit for current — shows up directly in welding (arc amperage) and in the heat input formula.
Amount of substance — mole (mol)	Rarely used directly in this course; alloy chemistry is reported by mass percent instead, but this is the base unit behind it in formal materials science.
Luminous intensity — candela (cd)	The base unit behind illuminance (lux) — relevant to the minimum lighting requirement for visual testing ($VT \geq 500$ lux).
Why only 4 of the 7 dominate this course	Nearly every equation in this glossary is built from just length, mass, time, and temperature. Recognizing that shrinks the whole unit system down to a handful of building blocks.

Dimensional Analysis — Building Derived Units

Derived unit (concept)	A unit for a quantity that is not one of the seven base quantities — built by combining base units the same way the physical formula combines the base quantities. If $\sigma = F/A$, then the unit for σ must be (unit of F)/(unit of A).
Force — newton: $N = kg \cdot m/s^2$	From Newton's second law, $F = m \cdot a$. Mass (kg) times acceleration (m/s^2). Every load in this course — cutting force F_c , hardness test load, Charpy hammer weight — is ultimately a force in newtons (or kgf, see below).
Stress / Pressure — pascal: $Pa = N/m^2 = kg/(m \cdot s^2)$	From $\sigma = F/A$ — force divided by area. This is the derived unit behind every stress and pressure value in the course: yield strength, UTS, PREN's implicit chloride environment, HIP pressure, hydrostatic test pressure — all reported in Pa, kPa, MPa, or GPa.
Energy / Work — joule: $J = N \cdot m = kg \cdot m^2/s^2$	From $W = F \cdot d$ (force times distance) or $E = mgh$ (the Charpy/Izod energy formula). This is the unit behind impact toughness, and — divided by volume — behind specific cutting energy (J/mm^3).
Power — watt: $W = J/s = kg \cdot m^2/s^3$	Energy per unit time. This is the unit behind cutting power $P_c = F_c \cdot V_c / (60,000 \cdot \eta)$ and behind electrical power in welding ($V \times A$).
Density — kg/m^3 (or g/cm^3)	Mass per unit volume — directly built from the two base units mass and length ³ . Used to compare families at a glance: aluminum ≈ 2.7 g/cm ³ , titanium ≈ 4.5 g/cm ³ , steel ≈ 7.85 g/cm ³ .
Frequency — hertz: $Hz = 1/s$	Cycles per second — the unit behind UT/ET sound and induction frequency (1–10 MHz for UT; 1 kHz–1 MHz for ET) and behind fatigue loading rate.
The general method	Write the defining equation for the quantity → substitute the base (or already-derived) units for each variable → simplify algebraically. The result is guaranteed correct as long as the defining equation is correct — this is why dimensional analysis works as a check on any formula, not just a way to build new ones.

Worked Examples — Checking a Formula's Units

Heat input: $HI = (V \cdot A \cdot 60) / (1000 \cdot \text{travel speed})$	$V \cdot A$ = watts = J/s. Multiplying by 60 (s/min) converts to J/min, matching the travel speed's per-minute basis. Dividing by (mm/min) leaves J/mm; dividing by 1000 converts J to kJ — confirming the stated result of kJ/mm.
Turning roughness: $Ra = f^2 / (8r)$	f (feed, mm) squared divided by r (nose radius, mm) leaves units of mm — matching Ra's reported units (usually converted to μm by $\times 1000$). If f or r were entered in the wrong unit, the result would be off by orders of magnitude — exactly why unit consistency is checked before trusting a calculated Ra.
Specific cutting energy: $u = F_c / (t_1 \cdot b)$	F_c in newtons, t_1 and b in mm — so $t_1 \cdot b$ is an area (mm^2). $\text{N}/\text{mm}^2 = \text{MPa}$, but specific cutting energy is conventionally reported as J/mm^3 once cutting distance is folded in — the units track exactly with the physical picture of 'energy consumed per unit volume of metal removed.'
Why hardness numbers are the exception	HRC, HB, and HV are empirical index numbers, not true derived SI units — HB and HV do reduce to a load-over-area (kgf/mm^2) inside the formula, but the numbers are calibrated against standardized indenters and loads, so a '60 HRC' cannot be dimensionally converted to '60 HB' the way MPa converts to psi. Cross-scale conversion requires an empirical lookup table, not algebra.
Dimensionless numbers	Some of the most important quantities in this course carry no units at all: carbon equivalent (CE), PREN, strain (ϵ), the strain-hardening exponent (n), notch sensitivity (q), and stress concentration factor (Kt) are all ratios of like quantities — their units cancel completely, which is exactly why they can be used to rank or compare materials regardless of specimen size.

Shop-Floor & U.S. Customary Units Still in Use

ksi / psi (stress, pressure)	1 ksi = 1000 psi = 6.895 MPa. Used constantly in AWS filler classification (E7018 = 70 ksi minimum tensile) and ASME/API pressure ratings alongside SI values.
ft·lbf (energy)	1 ft·lbf = 1.356 J. The U.S. customary unit for Charpy/Izod impact energy, reported alongside or instead of joules.
°F (temperature)	$^{\circ}\text{C} = (^{\circ}\text{F} - 32) / 1.8$. AWS D1.1 preheat/interpass tables (Table 4.4) list both $^{\circ}\text{F}$ and $^{\circ}\text{C}$ side by side — always confirm which one a spec is using before setting a preheat torch or oven.
in / mil (length)	1 in = 25.4 mm exactly. 1 mil = 0.001 in = 0.0254 mm — the standard unit for coating thickness and for corrosion rate (mils per year, MPY).
kgf (force / load)	1 kgf = 9.81 N. Hardness test loads (Brinell, Vickers) are conventionally specified in kgf even though the underlying physics is a force in newtons.
lux (illuminance)	The SI-derived unit for VT lighting requirements (≥ 500 lux, or per procedure) — built from candela and area, rarely discussed but directly specified in inspection procedures.
Rule of thumb for this course	When a number looks unfamiliar, check its unit before its magnitude — a '150' could be $^{\circ}\text{F}$ (preheat), HB (hardness), or MPa (a low stress) depending

	entirely on the unit attached.
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Quick Math Review — Skills for Using These Equations

Order of Operations

PEMDAS / GEMDAS	Parentheses/Grouping → Exponents → Multiplication & Division (left to right) → Addition & Subtraction (left to right). This order is not optional — evaluating a formula in the wrong sequence gives a wrong number that still looks plausible.
Worked example: $CE = C + \frac{Mn}{6} + \frac{(Cr+Mo+V)}{5} + \frac{(Ni+Cu)}{15}$	Each division happens first and only applies to its own term — Mn/6 divides Mn alone, not the running total. Compute all four terms separately, then add them last.
Common mistake: dividing the whole sum instead of one term	Writing $(C + Mn)/6$ instead of $C + Mn/6$ changes the answer completely — watch for which quantities are actually inside the denominator's 'reach.'
Grouping symbols change the order	In $HB = 2F / [\pi D(D - \sqrt{D^2 - d^2})]$, everything inside the brackets must be fully resolved — including the square root — before dividing 2F by it.
Nested operations, innermost first	Work from the innermost parentheses outward: in $D - \sqrt{D^2 - d^2}$, compute D^2 and d^2 first, subtract, then take the square root, then subtract from D.

Fractions & Ratios

A fraction is a ratio	a/b compares a to b. Stress $\sigma = F/A$ is literally 'force per unit area' — the fraction bar means 'for each unit of the denominator.'
Fractions with different units are still valid ratios	Fatigue ratio $= S_e \div UTS$ divides a stress by a stress, so the units cancel and the result is a pure (dimensionless) number — useful for comparing materials regardless of size.
Cross-multiplication solves a proportion	If $a/b = c/d$, then $a \cdot d = b \cdot c$. This is exactly how a unit conversion works: $1 \text{ in} / 25.4 \text{ mm} = x \text{ in} / 100 \text{ mm} \rightarrow x = 100/25.4$.
Multiplying by a conversion factor of 1	Any conversion factor (e.g., $25.4 \text{ mm}/1 \text{ in}$) equals 1 in value, so multiplying by it changes only the units, never the physical quantity — this is the arithmetic behind every unit conversion in the previous section.
Reciprocals flip a ratio	If $r = t_1/t_2$ (chip thickness ratio), then $t_2/t_1 = 1/r$ — useful when a formula or a data table reports the inverse of the ratio you need.

Percentages

Percent means 'per hundred'	$37\% = 37/100 = 0.37$. Convert to decimal before using in most formulas unless the formula is explicitly written to accept the number as-is (see below).
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Watch which formulas want a whole number vs. a decimal	PREN and CE use %Cr, %Mn, etc. as the whole number (e.g., 18 for '18% Cr'), not 0.18 — plug in exactly what the alloy spec lists. Strain as a percentage, by contrast, is usually converted to a decimal (2% strain = 0.02) before use in stress-strain equations.
Percent change formula	% change = (final – initial)/initial × 100%. This single pattern generates %EL (elongation), %RA (reduction in area), and %CW (percent cold work) — only the quantity being compared changes.
AISI/SAE carbon 'points' are hundredths of a percent, not whole percent	The last two digits of a steel grade (e.g., the '45' in 1045) mean 0.45% carbon — misreading this as '45% carbon' is a factor-of-100 error with a very different (impossible) steel.
Percentages don't simply add across different bases	10% of one alloy's Cr plus 5% of another's Mo cannot be added directly unless both percentages are of the same total mass — always check what the percentage is 'of' before combining values from different sources.

Exponents & Powers

Basic exponent rules	$x^a \cdot x^b = x^{a+b}$ (multiply, add exponents). $(x^a)^b = x^{ab}$ (power of a power, multiply exponents). $x^{-1} = 1/x$ (negative exponent means reciprocal).
Squares in area and geometry	d^2 and D^2 in the Brinell formula are just the diameters multiplied by themselves — always compute the square before combining with other terms per order of operations.
Fractional exponents in the Hollomon equation: $\sigma = K \cdot \epsilon^n$	n is typically a decimal between 0 and 1 (e.g., 0.15–0.25 for steel). Raising strain to a fractional power means flow stress rises more slowly than strain itself — a bigger n means the material keeps hardening (and resisting necking) longer.
Solving for an exponentiated variable: Taylor's tool life, $V_c \cdot T^n = C$	Isolate $T^n = C/V_c$, then undo the exponent by raising both sides to the power $1/n$: $T = (C/V_c)^{1/n}$. This 'undo with the reciprocal power' move works for any equation of the form $x^n = (\text{something})$.
Powers of a stress-intensity range: Paris' Law, $da/dN = C(\Delta K)^m$	ΔK is raised to a power m before multiplying by the constant C — a small change in ΔK produces a much larger change in crack growth rate whenever $m > 1$, which is typical (m is often 2–4 for metals).

Square Roots & Radicals

A square root undoes a square	$\sqrt{x}$ is the number that, multiplied by itself, gives x . Equivalently, $\sqrt{x} = x^{1/2}$ — a square root is just an exponent of one-half.
Radical of a product splits apart: $\sqrt{a \cdot b} = \sqrt{a} \cdot \sqrt{b}$	Used to read the skin depth formula $\delta = 503/\sqrt{f \cdot \mu_r \cdot \sigma}$ — you can evaluate $f \cdot \mu_r \cdot \sigma$ first and take one square root, or take three separate square roots and multiply; both give the same answer.
Solving for a variable trapped under a root: Hall-Petch, $\sigma_y = \sigma_0 + k/\sqrt{d}$	Isolate the root term: $k/\sqrt{d} = \sigma_y - \sigma_0$. Take the reciprocal of both sides, then square: $d = [k/(\sigma_y - \sigma_0)]^2$. Squaring is how you 'undo' a square root once it's alone on one side.

Order matters inside a radical	In $D - \sqrt{(D^2 - d^2)}$, the subtraction $D^2 - d^2$ must be completed before the square root is taken — $\sqrt{(D^2)} - \sqrt{(d^2)}$ is NOT the same thing and gives a wrong (and much larger) answer.
A negative number under an even root has no real answer	If a calculation under a square root ever comes out negative, that's a signal a value was plugged in wrong (e.g., d entered larger than D in the Brinell formula) — it is not a sign to just drop the negative sign.

Scientific Notation & Orders of Magnitude

Scientific notation: $a \times 10^n$	Used for numbers far from 1 in either direction — a unit cell ($\sim 0.3 \text{ nm} = 3 \times 10^{-10} \text{ m}$) or a fatigue life (10^7 cycles). a is between 1 and 10; n is the number of places the decimal point moves.
Multiplying/dividing in scientific notation	Multiply (or divide) the leading numbers normally, then add (or subtract) the exponents: $(2 \times 10^3) \times (3 \times 10^4) = 6 \times 10^7$.
Reading a logarithmic axis (the S-N curve)	Each gridline on a log-scale cycle axis represents $\times 10$, not '+ some fixed amount' — the visual distance from 10^3 to 10^4 cycles is drawn the same as from 10^6 to 10^7 , even though the second gap represents 900,000 more cycles than the first.
Why log scales are used for fatigue and grain size	Cycles to failure and grain diameters span many orders of magnitude (thousands to tens of millions of cycles; nanometers to millimeters) — a linear axis would squash the small values into an unreadable sliver, so a log scale is used to keep both ends legible.
Order-of-magnitude sanity checks	Before trusting a calculator answer, estimate the size by rounding each number to its leading digit and power of ten — if the calculator disagrees by more than a factor of ten from that estimate, a decimal point or exponent was likely entered wrong.

Rearranging Equations to Solve for an Unknown

The balance principle	An equation is a balance — whatever operation you perform on one side (add, subtract, multiply, divide, square, take a root) must be performed identically on the other side to keep it true.
Worked example: solve heat input for travel speed	$HI = (V \cdot A \cdot 60) / (1000 \cdot s) \rightarrow$ multiply both sides by $(1000 \cdot s)$: $HI \cdot 1000 \cdot s = V \cdot A \cdot 60 \rightarrow$ divide both sides by $(HI \cdot 1000)$: $s = (V \cdot A \cdot 60) / (HI \cdot 1000)$.
Worked example: solve turning roughness for feed	$Ra = f^2 / (8r) \rightarrow$ multiply both sides by $8r$: $8r \cdot Ra = f^2 \rightarrow$ take the square root of both sides: $f = \sqrt{8r \cdot Ra}$.
Worked example: solve percent cold work for final area	$\%CW = (A_0 - A_f) / A_0 \times 100 \rightarrow$ divide by 100, multiply by A_0 : $A_0 \cdot (\%CW / 100) = A_0 - A_f \rightarrow$ rearrange: $A_f = A_0 - A_0 \cdot (\%CW / 100) = A_0(1 - \%CW / 100)$.
Check the answer by substituting back in	After solving for an unknown, plug the result back into the original formula and confirm it reproduces the known value — the fastest way to catch an algebra mistake before it becomes a wrong shop measurement or a wrong exam answer.

Geometry & Right-Triangle Trigonometry

Basic Geometry Refresher

Angles around a point / a straight line	Angles on a straight line add to 180°; angles all the way around a point add to 360°. A right angle is 90°, marked with a small square in a diagram rather than a curved arc.
Complementary vs. supplementary angles	Complementary angles add to 90° (the two non-right angles in a right triangle are always complementary). Supplementary angles add to 180° (used when reading adjacent angles on a weld joint or a cut line).
Perimeter and circumference	Perimeter is the total distance around any shape. For a circle, circumference $C = \pi \cdot D = 2\pi \cdot r$, where D is diameter and r is radius — used to convert a bar's diameter into the length of stock consumed per revolution during turning.
Area of a circle	$A = \pi \cdot r^2 = \pi \cdot D^2/4$. This is the formula behind converting a measured rod or wire diameter into a cross-sectional area for a stress calculation ($\sigma = F/A$).
Area of a rectangle and a triangle	Rectangle: $A = \text{base} \times \text{height}$. Triangle: $A = \frac{1}{2} \times \text{base} \times \text{height}$. Used for quick sanity-check estimates of a part's cross-section before a more exact calculation.
The Pythagorean theorem	For any right triangle, $a^2 + b^2 = c^2$, where c is the hypotenuse (the side opposite the right angle) and a, b are the two legs. Rearranged forms: $c = \sqrt{a^2 + b^2}$, or $a = \sqrt{c^2 - b^2}$.
Worked example: Brinell hardness geometry	The Brinell formula's $D - \sqrt{D^2 - d^2}$ term is literally the Pythagorean theorem applied to the ball indenter: D is the ball diameter (the hypotenuse-like term), d is the diameter of the impression, and the difference gives the depth of penetration.

Right-Triangle Trigonometry — Sine, Cosine, Tangent

Labeling a right triangle	Pick one of the two non-right angles and call it θ . The hypotenuse is always the longest side, opposite the right angle. The 'opposite' side is across from θ ; the 'adjacent' side is the other leg, next to θ .
SOH-CAH-TOA	$\sin(\theta) = \text{opposite/hypotenuse}$. $\cos(\theta) = \text{adjacent/hypotenuse}$. $\tan(\theta) = \text{opposite/adjacent} = \sin(\theta)/\cos(\theta)$. These three ratios are fixed for a given angle no matter how large or small the triangle is.
Using sin, cos, tan to find a missing side	If θ and one side are known, multiply to find another side: $\text{opposite} = \text{hypotenuse} \times \sin(\theta)$; $\text{adjacent} = \text{hypotenuse} \times \cos(\theta)$; $\text{opposite} = \text{adjacent} \times \tan(\theta)$.
Using inverse trig to find a missing angle	If two sides are known, use the inverse functions to recover the angle: $\theta = \sin^{-1}(\text{opposite/hypotenuse}) = \cos^{-1}(\text{adjacent/hypotenuse}) = \tan^{-1}(\text{opposite/adjacent})$. Most calculators label these as $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$.
Common reference angles worth memorizing	30-60-90 and 45-45-90 triangles come up often as rules of thumb: $\sin(30^\circ) = 0.5$, $\sin(45^\circ) = \cos(45^\circ) \approx 0.707$, $\sin(60^\circ) \approx 0.866$. The 0.707 value in

	particular reappears throughout this course (see the fillet weld throat below).
Degrees vs. calculator mode	Every angle in this course is given in degrees, not radians — before using sin/cos/tan on a calculator, confirm it is set to 'DEG' mode, since the same keystrokes in 'RAD' mode return a completely different (wrong) number.

Course Applications — Where Trig Shows Up in the Shop

Fillet weld throat thickness	For a standard 45° fillet weld, throat = $0.707 \times \text{leg size (w)}$. The 0.707 factor is exactly $\cos(45^\circ)$ — the throat is the shortest distance from the weld root to the face, i.e., the side of a 45-45-90 triangle adjacent to the 45° angle formed by the two equal legs.
Angle-beam ultrasonic testing (UT) geometry	An angled UT probe sends sound along a slant path L into the part at a known refracted angle θ (commonly 45°, 60°, or 70°). The straight-line depth below the surface and the surface distance back to a reflector are both found with right-triangle trig once L and θ are known.
UT slant path from time of flight	$L = v_s \cdot t/2$, where v_s is the shear-wave velocity in the material and t is the round-trip travel time read off the flaw detector. Dividing by 2 accounts for the sound traveling to the reflector and back.
UT depth and surface distance from the slant path	Depth below the surface: $d = L \cdot \cos(\theta)$. Surface distance from the probe index point: $s = L \cdot \sin(\theta)$. θ is measured from the part surface's normal (perpendicular), so cos gives the 'straight down' component and sin gives the 'along the surface' component.
Worked example: 70° probe locating a flaw	Given $v_s = 3,230 \text{ m/s} = 3.23 \text{ mm}/\mu\text{s}$ and a measured travel time $t = 18.5 \mu\text{s}$: slant path $L = 3.23 \times 18.5/2 = 29.9 \text{ mm}$. Depth $d = L \cdot \cos(70^\circ) = 29.9 \times 0.342 \approx 10.2 \text{ mm}$ below the surface. Surface distance $s = L \cdot \sin(70^\circ) = 29.9 \times 0.940 \approx 28.1 \text{ mm}$ from the probe index point.
Draft and included angles in forming and machining	Draft angles on a die or mold, the included angle of a cutting tool tip, and the half-angle of a countersink are all just the θ in a right triangle relating a height (or depth) to a horizontal offset — the same sin/cos/tan relationships apply, just relabeled for the geometry at hand.

Engineering Equations & Key Concepts

Stress, Strain & Strengthening Relationships (Weeks 2, 7)

Engineering stress: $\sigma = F / A_0$	Internal force per unit original cross-sectional area. Units: Pa, MPa, ksi.
Engineering strain: $\epsilon = \Delta L / L_0$	Change in length relative to original length — dimensionless, or expressed as %.
Hooke's Law: $\sigma = E \cdot \epsilon$	Linear elastic region — E is the elastic (Young's) modulus, the slope of the stress-strain curve below the proportional limit.
0.2% offset method	Draw a line parallel to the elastic slope starting at $\epsilon = 0.002$; its intersection with the curve is the reported yield strength when no natural yield point

	exists (e.g., aluminum).
$\%EL = (L_f - L_0)/L_0 \times 100\%$ · $\%RA = (A_0 - A_f)/A_0 \times 100\%$	Elongation at fracture and reduction in area — the two standard ductility measures from a tensile test.
True stress-strain (Hollomon equation): $\sigma = K \cdot \epsilon^n$	Flow stress during plastic deformation beyond yield. K = strength coefficient, n = strain-hardening exponent — both material constants used throughout forming and machining analysis.
True strain: $\epsilon_{true} = \ln(l/l_0)$	Unlike engineering strain, true strain accounts for the continuously changing gauge length/thickness — the correct measure once necking or heavy forming begins.
Hall-Petch relationship: $\sigma_y = \sigma_0 + k / \sqrt{d}$	Yield strength increases as grain size (d) decreases — the practical basis for grain-refinement strengthening.
Taylor (dislocation) hardening: $\sigma \propto \sqrt{\rho}$	Strength scales with the square root of dislocation density — the mechanism behind work hardening.
Percent cold work: $\%CW = (A_0 - A_f)/A_0 \times 100\%$	Same form as reduction in area — quantifies how much a part has been cold-worked, which sets its strength/ductility per the %CW-vs-property curves and its recrystallization temperature on annealing.

Stress Concentration, Notches & Fatigue Design (cross-cutting — Weeks 2, 9, 11, 12, 15)

Stress concentration factor: $K_t = \sigma_{max} / \sigma_{nom}$	The theoretical (geometric) ratio by which a notch, hole, fillet, undercut, or sharp corner locally amplifies nominal applied stress — independent of material, a function of geometry alone (found from Peterson's charts or FEA).
Notch sensitivity: $q = (K_f - 1) / (K_t - 1)$	How much of the theoretical stress concentration a real material actually 'feels' in fatigue (0 = insensitive, 1 = fully sensitive). Brittle/high-strength materials tend toward $q \approx 1$; soft, ductile materials can locally yield and blunt the notch, giving $q < 1$.
Fatigue notch factor: $K_f = 1 + q(K_t - 1)$	The effective stress concentration factor for fatigue life prediction — always $\leq K_t$.
Real-world K_t magnitudes	Weld toes/roots: $K_t \approx 2-5$ (a major reason HIC and fatigue cracks concentrate there). Turbine cooling holes: $K_t \approx 2.8$. Even a fillet, keyway, or machining mark can produce a locally significant K_t — this is why 'the weld meets the drawing' does not guarantee fatigue performance.
Common manufacturing stress risers	Undercut, weld toe/root, sharp fillet radius, keyway corner, thread root, tool marks/chatter, porosity, inclusions, sudden section-thickness change — each one is a K_t source introduced (or avoided) by process control.
Residual stress: compressive vs. tensile	Compressive surface residual stress (shot peening, down/climb milling, cold rolling) delays fatigue crack initiation; tensile residual stress (grinding burn, welding, EDM recast layer) accelerates it — same applied load, very different fatigue life depending on which sign is present at the surface.
Mitigation methods	Generous fillet radii, smooth transitions, post-weld toe grinding or TIG

	dressing, shot peening, stress-relief PWHT — all exist specifically to lower Kt or convert tensile residual stress to compressive at the highest-Kt location.
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Hardness Testing Formulas (Week 7 + Hardness Scales reference)

Rockwell: $HR = N - (h / s)$	N = scale constant, h = permanent depth increase (mm) after removing the major load, s = 0.002 mm/division. Higher number = shallower indent = harder material.
Brinell: $HB = 2F / [\pi D (D - \sqrt{D^2 - d^2})]$	F = load (kgf), D = ball diameter (mm), d = indentation diameter (mm). Standard test: 10 mm ball, F/D ² held constant (30 for steel).
Vickers: $HV = 1.854 \times F / d^2$	F = load (kgf), d = mean diagonal of the square indentation (mm). One continuous scale from soft metals to diamond-like coatings.
UTS approximation from Brinell: $UTS \text{ (MPa)} \approx 3.45 \times HB$	Empirical, carbon/low-alloy steels only, valid roughly HB 100–400 — do not use for stainless or non-ferrous alloys.

Impact, Fatigue & Fracture Mechanics (Week 7 + Impact/Fatigue reference)

Charpy/Izod absorbed energy: $E = mg(h_0 - h_1)$	m = hammer mass, g = 9.81 m/s ² , h ₀ = release height, h ₁ = height reached after fracture. Reported in joules or ft-lb — higher = tougher.
Endurance limit approximation: $S_e \approx 0.5 \times UTS$	For polished, unnotched steel specimens with UTS below ~1400 MPa. Above that, inclusions dominate and the ratio breaks down. Note this is the smooth-specimen value — apply Kf above to estimate the notched/as-manufactured endurance limit.
Fatigue ratio = $S_e \text{ (or fatigue limit)} \div UTS$	Typically 0.4–0.5 for steels, 0.3–0.4 for aluminum alloys — useful for comparing materials at a glance.
S-N (Wöhler) curve	Stress amplitude vs. log(cycles to failure). Steels show a true horizontal endurance limit; aluminum and most non-ferrous alloys do not and must specify a life (e.g., 10 ⁷ cycles).
Paris' Law: $da/dN = C(\Delta K)^m$	Governs Stage II fatigue crack growth rate per cycle as a function of the stress intensity factor range ΔK ; C and m are material constants — this is the crack-growth counterpart to the Kt/Kf stress-concentration concepts above.

Manufacturing — Cutting Mechanics, Tool Life & Machining Parameters (Week 12 + Cutting/Turning-Milling references)

Cutting speed / spindle speed: $V_c = \pi \cdot D \cdot n / 1000$ $n = 1000 \cdot V_c / (\pi \cdot D)$	D = workpiece or cutter diameter (mm), n = spindle speed (rpm). The most fundamental machining relationship — used to set up every turning and milling operation.
Material removal rate: $MRR = a_p \cdot f \cdot V_c$	a _p = depth of cut, f = feed, v _f = table feed rate, a _e = radial engagement — the productivity metric that cutting parameter selection ultimately optimizes.

(turning) · MRR = $a_e \cdot a_p \cdot v_f$ / 1000 (milling)	
Merchant's shear angle: $\phi = 45^\circ + \alpha/2 - \lambda/2$	α = tool rake angle, λ = friction angle (= $\arctan \mu$). Higher rake or lower friction → larger shear angle → thinner chip → less cutting energy.
Chip thickness ratio: $r = t_1/t_2 = \sin \phi / \cos(\phi - \alpha)$	t_1 = uncut chip thickness, t_2 = actual chip thickness. Always $r < 1$ since the chip is thicker than the uncut layer (it must be, by volume conservation, once it shears and curls).
Specific cutting energy: $u = F_c / (t_1 \cdot b)$	F_c = cutting force, b = width of cut, in J/mm ³ . Depends on work material hardness and chip thickness (the 'size effect') — power = $u \times \text{MRR}$.
Cutting force (Merchant's circle): $F_c = \tau_s \cdot A_s / [\sin \phi \cdot \cos(\lambda - \alpha) / \cos(\phi + \lambda - \alpha)]$	τ_s = shear yield stress, A_s = shear-plane area = $t_1 \cdot b / \sin \phi$. The thrust force component F_t acts perpendicular to V_c and can even reverse sign (pulling the tool into the work) at high positive rake.
Taylor's tool life equation: $V_c \cdot T^n = C$	T = tool life (min), n = Taylor exponent (0.1–0.4 HSS; 0.2–0.5 carbide), C = constant for the tool-material pair. Cutting speed is the single strongest driver of tool life — a modest V_c increase costs tool life exponentially.
Turning surface roughness: $R_a \approx f^2 / (8 \cdot r_\epsilon)$	f = feed per revolution, r_ϵ = tool nose radius. Finer feed and larger nose radius give a smoother surface — the same relationship introduced in Week 12, restated here alongside its milling counterpart.
Milling surface roughness: $R_a \approx f_z^2 / (8 \cdot r_\epsilon)$ (flat end) · $R_a \approx a_e^2 / (8R)$ (ball end, scallop height)	f_z = feed per tooth, R = ball radius, a_e = radial step-over. Governs the theoretical finish achievable from tool geometry and feed alone, before chatter or tool wear are considered.
Chip thinning (milling): $hm = f_z \cdot \sqrt{a_e / D_c}$	When radial engagement a_e is less than half the cutter diameter D_c , the actual chip is thinner than the programmed feed per tooth — feed must be increased ($f_{z_adj} = f_{z_rec} \cdot \sqrt{D_c / a_e}$) to restore the intended chip load and avoid rubbing.
Cutting power: $P_c = F_c \cdot V_c / (60,000 \cdot \eta)$	η ≈ machine efficiency (~0.80). Used to verify a cutting parameter set against available spindle motor power before committing to a program.
Heat partition in cutting	Roughly 75–80% of cutting heat leaves in the chip — why chip control and coolant matter for workpiece surface integrity (and why insufficient chip evacuation drives tool wear and workpiece thermal damage).

Manufacturing — Forming, Drawing & Sheet Metal (Weeks 10, 11 + Extrusion/FLD references)

Extrusion ratio: $ER = A_0 / A_f$	Billet cross-sectional area divided by extrudate cross-sectional area. Typical range 10:1 to 100:1 for aluminum — a higher ratio demands more extrusion pressure and favors more extrudable (lower-strength) alloy series.
Recrystallization temperature: $T_{rex} \approx 0.4 \cdot T_m$	T_m = absolute melting point. The dividing line between hot working (no net work hardening) and cold working (work hardening accumulates) for a given metal.
Strain-hardening exponent n and strength coefficient	From $\sigma = K \cdot \epsilon^n$. Higher n means a material spreads strain more uniformly before localizing into a neck — directly predicts formability in stretching

K	operations.
Plastic strain ratio (Lankford coefficient): $\bar{r}$	Ratio of width strain to thickness strain in a tensile test — measures resistance to thinning. Higher $\bar{r}$ (e.g., ferritic stainless, some aluminum) means better deep-drawability.
Forming Limit Curve (FLC₀) estimate: $FLC_0 \approx (23.3 + 360 \cdot t) \cdot n / 0.21$	t = sheet thickness (mm), n = strain-hardening exponent. Gives the plane-strain forming limit — the minimum point of the FLC and the critical value for most stamping operations.
Forming Limit Diagram (FLD) axes: major strain ϵ_1 (y-axis) vs. minor strain ϵ_2 (x-axis)	Both are true strains, $\epsilon = \ln(l/l_0)$. Points below the FLC are safe; points above indicate through-thickness necking/fracture; points in the lower-left region ($\epsilon_2 < -\epsilon_1/2$) risk wrinkling instead.
Draw/reduction limits per pass	Wire/rod drawing: 15–25% area reduction per die. Tube drawing: 15–45% depending on method (sinking, fixed plug, floating plug, rod/mandrel) — exceeding these limits risks central bursts (chevron cracking) from excessive hydrostatic tension in the deformation zone.
Springback	Elastic recovery after a bend or draw releases — the part relaxes toward its original shape by an amount that depends on E , yield strength, and bend radius/thickness ratio; tooling must overbend to compensate.

Weldability & Heat Input (Weeks 4, 14, 15 + Preheat/ESAB references)

IIW Carbon Equivalent: $CE = C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15$	The primary weldability index in AWS D1.1 — higher CE means higher hardenability and higher required preheat.
Pcm (Ito-Bessyo): $Pcm = C + Si/30 + (Mn+Cu+Cr)/20 + Ni/60 + Mo/15 + V/10 + 5B$	Better than CE for low-carbon, micro-alloyed/HSLA steels (<0.18% C) — gives a more accurate preheat estimate for those grades.
Heat Input: $HI = (V \times A \times 60) / (1000 \times \text{travel speed}) \rightarrow \text{kJ/mm}$	Higher heat input $\rightarrow$ slower cooling $\rightarrow$ coarser grains, wider HAZ, more distortion. Lower heat input $\rightarrow$ faster cooling $\rightarrow$ finer grains but higher HIC risk in hardenable steel.
Preheat threshold bands by CE	$CE < 0.40$: generally no preheat. $CE 0.40\text{--}0.60$: preheat $\sim 100\text{--}200^\circ\text{C}$. $CE > 0.60$: preheat $200\text{--}300^\circ\text{C}+$, mandatory for most applications.
Dilution by process	SMAW 20–30% · GMAW 25–35% · SAW 50–70% — the fraction of deposited weld metal that is melted base metal; the WPS accounts for this when specifying filler composition.
Schaeffler diagram — $Ni_{eq} = Ni + 30C + 0.5Mn$	Nickel equivalent (austenite formers) used with a chromium equivalent to predict resulting phase balance (ferrite/austenite/martensite) in stainless steel weld metal.
WRC-1992 — $Cr_{eq} = Cr + Mo + 0.7Nb$, $Ni_{eq} = Ni + 35C + 20N + 0.25Cu$	A more accurate modern update to the Schaeffler diagram for duplex and modern stainless weld metal; predicts Ferrite Number (FN) — target 4–8 FN for hot-crack resistance without embrittlement.

Corrosion & Electrochemistry (Week 16)

PREN = %Cr + 3.3 × %Mo + 16 × %N	Pitting Resistance Equivalent Number — ranks stainless/nickel alloys by chloride pitting resistance. 304 ≈ 19, 316 ≈ 24, 2205 duplex ≈ 35–40.
PREN thresholds	<18 low resistance · 18–25 moderate (fresh water) · 25–35 good (brackish/marine) · 35–40 high (seawater) · >40 very high (full seawater/desalination).
Cathodic protection criterion: –0.85 V vs. Cu/CuSO₄	Per NACE SP0169 — the minimum structure-to-electrolyte potential considered adequately protected.
Corrosion rate: mm/year or mils per year (MPY)	The standard way uniform corrosion rate is measured and used to calculate a corrosion allowance and remaining service life.

NDT Physics (Week 8 + NDT Methods reference)

Eddy current skin depth: δ = $503 / \sqrt{f \times \mu_r \times \sigma}$	f = frequency (Hz), μ_r = relative magnetic permeability, σ = conductivity (MS/m). Higher frequency = shallower, more surface-sensitive inspection; lower frequency reaches deeper but loses surface sensitivity.
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